Date
16 October 2026

Project Name	Metro Ticket Generating System in ServiceNow	**Maximum Marks**	4 Marks

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	**User Authentication & Access Control**	Log in using ServiceNow credentials with Administrator/Developer access. Role-based access control for Commuters, Approvers, and Admins.
FR-2	**Master Data Configuration**	Configure master station lists and route mapping tables. Define fare rules matrix based on distance/zones. Manage approval groups and transport administrator groups.
FR-3	**Service Portal Ticket Booking**	Catalog form interface to capture Source & Destination stations. Capture Travel Date, Ticket Type (Single, Return, Pass), and Passenger Details. Client-side script validation preventing identical source/destination selection.
FR-4	**Dynamic Fare Calculation**	Calculate fare amount, route distance, and ticket summary dynamically using Script Includes. Real-time price recalculation upon updating passenger quantity or ticket type.
FR-5	**Automated Approval Workflow**	Route ticket request to line manager or transport admin via Flow Designer. Trigger automated email notifications to approvers with quick approval links.
FR-6	**Ticket Generation & Delivery**	Auto-generate unique ticket number and digital pass upon request approval. Send email notification containing generated ticket summary and PDF/QR code. Update ticket state (Pending → Approved → Generated).
FR-7	**Status Tracking & Cancellation**	Track ticket status from Service Portal "My Requests" dashboard. Execute automated refund or fare adjustment logic on ticket cancellation via Scheduled Jobs.

PROJECT DESIGN PHASE-II

Solution Requirements (Functional & Non-functional)

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	**Usability**	The Service Portal interface must be responsive, intuitive, and accessible on both desktop and mobile web browsers with maximum 3-click ticket selection.
NFR-2	**Security**	Ensure strict ServiceNow ACLs (Access Control Lists) to prevent unauthorized access to master fare tables and passenger personal identity data.
NFR-3	**Reliability**	The system must guarantee 99.9% accuracy in automated fare calculation and ensure zero duplicate ticket number generation during concurrent user requests.
NFR-4	**Performance**	Catalog client scripts and Script Includes must process and update fare summaries within 1.5 seconds of user input on the Service Portal form.
NFR-5	**Availability**	The ticket booking portal must maintain 99.9% uptime operating on the cloud ServiceNow instance, ensuring 24/7 booking access for commuters.
NFR-6	**Scalability**	The system logic and Flow Designer workflows must support concurrent ticket bookings across multiple urban metro lines without performance degradation.

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